

SIDO KANHU MURMU UNIVERSITY DUMKA



NEP-2020 BASED COURSE CURRICULUM (ZOOLOGY)

For
UNDERGRADUATE
PROGRAMME

[B.Sc. (Major, Minor and Advance Major)]

ACADEMIC SESSION

w.e.f. 2023-2027

29/04/24
Dr. Nilesh Kumar
(HOD, P.G. Zoology)

29.04.2024
Dr. Nirmala Tripathi
(Asst. Prof.)

29.4.24
Dr. Punam Hembrom
(Asst. Prof.)

29 April 2024
Dr. Amita Hembrom
(Asst Prof.)

29.4.24
Dr. Khushboo Tigga
(Asst. Prof.)

(Dr. Parimal Khan)
External(Patna University)

29/04/2024
Dean, Faculty
Group

(Dr. T.P. Singh)
Member S.B.O.S,
NEP-2020

Members Of Board of Studies for Preparing Provisional Syllabus of The Four-Year Undergraduate Programme (FYUGP)

CHAIRMAN:

Dr. Nilesh Kumar

HOD & Associate Professor, University Department of Zoology, S.K.M.U., Dumka

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DEAN SCIENCE:

1. Dr. S.N. Adhikary

Associate Professor, University Department of Mathematics, S.K.M.U.

HIGHLIGHTS OF REGULATIONS OF FYUGP

PROGRAMME DURATION

- The Full-time, Regular UG programme for a regular student shall be for a period of four years with multiple entry and multiple exit options.
- The session shall commence from **1st of July**.

ELIGIBILITY

- The selection for admission will be primarily based on availability of seats in the Major subject and marks imposed by the institution. Merit point for selection will be based on marks obtained in Major subject at Class 12 (or equivalent level) or the aggregate marks of Class 12 (or equivalent level) if Marks of the Major subject is not available. Reservation norms of The Government of Jharkhand must be followed as amended in times.
- UG Degree Programmes with Double Major shall be provided only to those students who secure a minimum of overall 75% marks (7.5 CGPA) or higher.
- Other eligibility criteria including those for multiple entry will be in light of the UGC Guidelines for Multiple Entry and Exit in Academic Programmes offered in Higher Education Institutions.

ADMISSION PROCEDURE

- The reservation policy of the Government of Jharkhand shall apply in admission and the benefit of the same shall be given to the candidates belonging to the State of Jharkhand only. The candidates of other states in the reserved category shall be treated as General category candidates. Other relaxations or reservations shall be applicable as per the prevailing guidelines of the University for FYUGP.

VALIDITY OF REGISTRATION

- Validity of a registration for FYUGP will be for maximum for Seven years from the date of registration.

ACADEMIC CALENDAR

- An Academic Calendar will be prepared by the university to maintain uniformity in the CBCS of the UG Honours Programmes, UG Programmes, semesters and courses in the college run under the university (Constituent/Affiliated).
- Academic Year: Two consecutive (one odd + one even) semesters constitute one academic year.
- Semester: The Odd Semester is scheduled from July to December and the Even Semester is from January to June. Each week has a minimum of 40 working hours spread over 6 days.
- Each semester will include – Admission, course work, conduct of examination and declaration of results including semester break.
- In order to undergo 8 weeks' summer internship/ apprenticeship during the summer camp, the Academic Calendar may be scheduled for academic activities as below:
 - a) **Odd Semester:** From first Monday of August to third Saturday of December
 - b) **Even Semester:** From first Monday of January to third Saturday of May
- An academic year comprising 180 working days in the least is divided into two semesters, each semester having at least 90 working days. With six working days in a week, this would mean that each semester will have $90/6 = 15$ teaching/ working weeks. Each working week will have 40 hours of instructional time.
- Each year the University shall draw out a calendar of academic and associated activities, which shall be strictly adhered to. The same is non-negotiable. Further, the Department will make all reasonable to deliver the programmes of study and other educational services as mentioned in its Information Brochure and website. However, circumstances may change prompting the Department to reserve the right to change

the content and delivery of courses, discontinue or combine courses and introduce or withdraw areas of specialization.

PROGRAMME OVERVIEW/ SCHEME OF THE PROGRAMME

Undergraduate degree programmes of either 3 or 4-year duration, with multiple entries and exit points and re-entry options within this period, with appropriate certifications such as:

- UG Certificate after completing 1 year (2 semesters) of study in the chosen fields of study provided they complete one vocational course of 4 credits during the summer vacation of the first year or internship/ Apprenticeship in addition to 6 credits from skill-based courses earned during first and second semester.,
- UG Diploma after 2 years (4 semesters) of study diploma provided they complete one vocational course of 4 credits or internship/ Apprenticeship/ skill based vocational courses offered during first year or second year summer term in addition to 9 credits from skill-based courses earned during first, second, and third semester,
 - a) Bachelor's Degree after a 3-year (6 semesters) programme of study,
 - b) Bachelor's Degree (Honours) after a 4-year (8 semesters) programme of study.
 - c) Bachelor Degree (Honours with Research) after a 4-year (8 semesters) programme of study to the students undertaking 12 credit Research component in fourth year of FYUGP.

CREDIT OF COURSES

The term 'credit' refers to the weightage given to a course, usually in terms of the number of instructional hours per week assigned to it. The workload relating to a course is measured in terms of credit hours. It determines the number of hours of instruction required per week over the duration of a semester (minimum 15 weeks).

- a) One hour of teaching/ lecture or two hours of laboratory /practical work will be assigned per class/interaction.

One credit for Theory = 15 Hours of Teaching i.e., 15 Credit Hours

One credit for Practicum = 30 Hours of Practical work i.e., 30 Credit Hours

- b) For credit determination, instruction is divided into three major components:

Lectures (L) – Classroom Hours of one-hour duration. **Tutorials (T)** – Special, elaborate instructions on specific topics of one-hour duration. **Practical (P)** – Laboratory or field exercises in which the student has to do experiments or other practical work of two-hour duration.

CALCULATION OF MARKS FOR THE PURPOSE OF RESULT

- Student's final marks and the result will be based on the marks obtained in Semester Internal Examination and End Semester Examination organized taken together.
- Passing in a subject will depend on the collective marks obtained in Semester internal and End Semester University Examination both. However, students must pass in Theory and Practical Examinations separately.

PROMOTION CRITERIA

First degree programme with single major:

- i. The Requisite Marks obtained by a student in a particular subject will be the criteria for promotion to the next Semester.
- ii. No student will be detained in odd Semesters (I, III, V & VII).
- iii. To get promotion from Semester-II to Semester-III a student will be required to pass in at least 75% of Courses in an academic year, a student has to pass in minimum 9 papers out of the total 12 papers.
- iv. To get promotion from Semester-IV to Semester-V (taken together of Semester I, II, III & IV) a student has to pass in minimum 18 papers out of the total 24 papers.

- v. To get promotion from Semester-VI to Semester-VII (taken all together of Semester I, II, III, IV, V & VI) a student has to pass in minimum 26 papers out of the total 34 papers.
- vi. However, it will be necessary to procure pass marks in each of the paper before completion of the course.

First degree programme with dual major:

- vii. Above criteria are applicable as well on the students pursuing dual degree programmes however first-degree programme will remain independent of the performance of the student in dual major courses.
- viii. To get eligible for taking ESE, a student will be required to pass in at least 75% of Courses in an academic year.
- ix. A student has to pass in minimum 3 papers out of the total 4 papers.
- x. It will be a necessity to clear all papers of second major programme in second attempt in succeeding session, failing which the provision of dual major will be withdrawn and the student will be entitled for single first-degree programme.

UG Degree Programmes with Single Major:

- A student has to secure a minimum of 50% credits from the major discipline for the 3-year/4-year UG degree to be awarded a single major. For example, in a 3-year UG programme, if the total number of credits to be earned is 120, a student of Physics with a minimum of 60 credits will be awarded a B.Sc. in Physics with a single major. Similarly, in a 4-year UG programme, if the total number of credits to be earned is 160, a student of Physics with a minimum of 80 credits will be awarded a B.Sc. (Hons./Hon. With Research) in Physics in a 4-year UG programme with single major.

UG Degree Programmes with Double Major:

- A student has to secure a minimum of 40% credits from the second major discipline for the 3-year/4-year UG degree to be awarded a double major. For example, in a 3-year UG programme, if the total number of credits to be earned is 120, a student of Physics with a minimum of 48 credits will be awarded a B.Sc. in Physics with a double major. Similarly, in a 4-year UG programme, if the total number of credits to be earned is 160, a student of Physics with a minimum of 64 credits will be awarded a B.Sc. (Hons. with Research) in Physics in a 4-year UG programme with double major.

Interdisciplinary UG Programmes:

- The credits for core courses shall be distributed among the constituent disciplines/subjects so as to get core competence in the interdisciplinary programme. For example, a degree in Econometrics requires courses in economics, statistics, and mathematics. The total credits to core courses shall be distributed so that the student gets full competence in Econometrics upon completion of the programme. The degree for such students will be awarded as B.Sc. in Econometrics for a 3-year UG programme or B.Sc. (Honours) / B.Sc. (Honours with Research) in Econometrics for a 4-year UG programme.

Multidisciplinary UC Programmes:

- In the case of students pursuing a multidisciplinary programme of study, the credits to core courses will be distributed among the broad disciplines such as Life sciences, Physical Sciences, Mathematical and Computer Sciences, Data Analysis, Social Sciences, Humanities, etc., For example, a student who opts for a

UG program in Life sciences will have the total credits to core courses distributed across Botany, Zoology and Human biology disciplines. The degree will be awarded as B.Sc. in Life Sciences for a 3-year programme and B.Sc. (Honours) in Life Sciences or B.Sc. (Honours with Research) for a 4-year programme without or with a research component respectively. The statutory bodies of the Universities and Colleges such as the Board of Studies and Academic Council will decide on the list of courses under major category and credit distribution for double major, interdisciplinary and multidisciplinary programmes.

Major Courses: (Disciplinary / Interdisciplinary Major)

- The **FYUGP** offers students a general understanding as well as in-depth study of at least one subject area termed as Major.
- At the time of admission, a student will select a Major subject offered by the institution.
- The major subject will be of a higher level to get deep knowledge and specialization in the discipline selected. These will be of total 80 credits to be covered during semesters I -VIII.
- In semesters VII and VIII, the advanced level courses of the major subject of 12 credits be offered in lieu of taking up Research Courses.

Minor Courses: (Disciplinary/Interdisciplinary Minor)

- The student will have to choose Two Minor subjects at the time of admission.
- One Minor subject to support the major subject from the concerned broad disciplinary area in the first semester.
- The other will be from a pool of vocational courses/skill enhancement courses of vocational nature either supporting the major or from the field of major courses offered by the institution.

Common Courses:

The following courses may be common to all students in semesters I to III:

- A student has to opt for One paper of 2 credits in MIL (Hindi Communication).
- English Communication will be a compulsory common paper of 2 credits.
- 'Environmental studies' of 2 credits
- 'Understanding India' of 2 credits
- 'Health & Wellness, Yoga Education, Sports & Fitness' of 2 credits
- 'Community Engagement & Service (NSS/ NCC/ Adult Education)' of 2 credits

Table 1: Credit Framework for Four Year Undergraduate Programme (FYUGP) under State Universities of Jharkhand
[Total Credits = 160]

Level of course	Semester	MJ; Discipline Specific Courses - Core or Major (80)	MN; Minor from discipline (16)	MN; Minor from vocational (16)	MDC; Multidisciplinary Courses [Life sciences, Physical Sciences, Mathematical and Computer Sciences, Data Analysis, Social Sciences, Humanities. etc.] (9)	AEC; Ability Enhancement Courses (Modem Indian Language and English) (8)	SEC; Skill Enhancement Courses (9)	VAC; Value Added Courses (6)	IAP; Internship Dissertation (4)	RC; Research Courses (12)	AMJ; Advanced Courses in lieu of Research (12)	Credits	Double Major (DMJ)
1	2	3	4	5	6	7	8	9	10	11	12	13	14
100-199: Foundation or Introductory Courses	I	4	4		3	2	3	4 (2+2)				20	4+4
	II	4+4		4	3	2	3					20	4+4
	Exit Point: Undergraduate Certificate provided with Summer Internship Project (4 credits)												
200-299: Intermediate Level Courses	III	4+4	4		3	2	3					20	4+4
	IV	4+4+4		4		2		2				20	4+4
	Exit Point: Undergraduate Diploma provided with Summer Internship In I" or 2" year/ Project (4 credits)												
300-399: Higher Level Courses	V	4+4+4	4						4			20	4+4
	VI	4+4+4+4		4								20	4+4
	Exit Point: Bachelor's Degree												
400-499: Advanced Courses	VII	4+4+4+4	4									20	4+4
	VIII	4		4						12	4+4+4	20	4+4
	Exit Point: Bachelor's Degree with Hons. Hons. with Research											160	224

Note: Honours students not undertaking research will do 3 courses for 12 credits in lieu of a Research project / Dissertation.

COURSE STUCTURE FOR FYUGP ‘ZOOLOGY HONOURS/ RESEARCH’

Table 1: Credit Framework for Four Year Undergraduate Programme (FYUGP) under State Universities of Jharkhand [Total Credits = 160]

Semester	Paper Code	Papers	Credits	Marks distribution in examination		
				Mid SEM (FM)	End SEM Theory (FM)	End SEM Practica l/ Viva (FM)
FIRST YEAR						
SEM I	MJ-Zoo-01	Non-chordates	4	25	75	-
	MN-01	Paper -1 from table -2	4	-	75	25
	MDC-1	To be chosen from table-2 other disciplines	3	-	75	-
	AEC-1	Hindi / English	2	-	50	-
	VAC-1	Environmental studies & Understanding India	2+2	-	50+50	-
	SEC-1	Digital education	3	-	75	-
SEM II	MJ-Zoo-02	Chordates and comparative anatomy of vertebrates	4	25	75	-
	MJ- Zoo-03	Practical: Based upon MJ-Zoo-01 and MJ-Zoo-02	4	25	-	75
	MN-VOC-1	Paper-1 from table-2 selected pool of vocational course	4	25	75	-
	MDC-2	To be chosen from table-2 other disciplines	3	-	75	-
	AEC-2	Hindi / English	2	-	50	-
	SEC-2	Mathematical and Computational thinking Analysis	3	-	75	-
SECOND YEAR						
SEM III	MJ- Zoo-04	Osteology, Evolution, Economic Zoology and Animal Behaviour	4	25	75	-
	MJ- Zoo-05	Practical: Based upon MJ-Zoo-04	4	25		75
	MN-02	Paper -2 of selected group in SEM-I from table -2	4	-	75	25
	MDC-3	To be chosen from table-2 other disciplines	3	-	75	-
	AEC-3	Any other Language Course except Hindi and English (paper-I)	2	-	50	-
	SEC-3	Spoken English	3	-	75	-
	MJ- Zoo-06	Physiology, Endocrinology and Reproductive Biology	4	25	75	-

SEM IV	MJ-Zoo-07	Biochemistry, Tools and Techniques	4	25	75	-
	MJ- Zoo-08	Practical: Based upon MJ-Zoo-06 and MJ-Zoo-07	4	25	-	75
	MN-VOC-2	Paper-2 of selected vocational courses in SEM-I	4	25	75	-
	AEC-4	Any other Language Course except Hindi & English (paper-II)	2	-	50	-
	VAC-2	Health & wellness, Yoga Education, Sports & fitness	2	-	50	-
THIRD YEAR						
SEM V	MJ- Zoo-09	Cell Biology and Molecular Biology	4	25	75	-
	MJ-Zoo-10	Genetics	4	25	75	-
	MJ-Zoo-11	Practical: Based upon MJ-Zoo-09 and MJ-Zoo-10	4	25	-	75
	MN-03		4	-	75	25
	IAP		4	25	75	-
SEM VI	MJ-Zoo-12	Immunology and Parasitology	4	25	75	-
	MJ-Zoo-13	Developmental Biology	4	25	75	-
	MJ-Zoo-14	Ecology and Toxicology	4	25	75	-
	MJ-Zoo-15	Practical: Based upon MJ-Zoo-12, MJ-Zoo-13 and MJ-Zoo-14	4	25	-	75
	MN-VOC-3		4	25	75	-
FOURTH YEAR						
SEM VII	MJ-Zoo-16	Microbiology	4	25	75	-
	MJ-Zoo-17	Biotechnology	4	25	75	-
	MJ-Zoo-18	Biostatistics, Histology and Histochemistry	4	25	75	-
	MJ-Zoo-19	Practical: Based upon MJ-Zoo-16, MJ-Zoo-17 and MJ-Zoo-18	4	25	-	75
	MN-04		4	-	75	25
B.Sc. Zoology (Honours)						
SEM VIII	MJ-Zoo-20	Seminar (Presentation and Discussion)	4	-	-	100
	MN-VOC-4		4	25	75	-
	AMJ-Zoo-01	Environmental studies	4	25	75	-
	AMJ-Zoo-02	Hematology	4	25	75	-
	AMJ-Zoo-03	Practical	4	25	-	75
B.Sc. Zoology (Honours With research)						
SEM VIII	MJ-Zoo-20	Seminar (Presentation and Discussion)	4	-	-	100
	MN-VOC-4		4	25	75	-
	RC	Research methodology	4	25	75	-
		Research Project/ Dissertation/ Research internship/ Field work	8	-	-	200
Total Credit			160			

Note: Honours students not undertaking research will do 3 courses for 12 credits in lieu of a Research project / Dissertation.

Pass Mark will be 40% for Mid Sem (Internal) and End Sem (External) separately.

Abbreviations:

AEC	Ability Enhancement Courses
SEC	Skill Enhancement Courses
IAP	Internship/Apprenticeship/ Project
MDC	Multidisciplinary Courses
MJ	Major Disciplinary/Interdisciplinary Courses
DMJ	Double Major Disciplinary/Interdisciplinary Courses
MN	Minor Disciplinary/Interdisciplinary Courses
AMJ	Advanced Major Disciplinary/Interdisciplinary Courses
RC	Research Courses

Table 2: Subject combination chart for UG ZOOLOGY FYGP as per NEP-2020

MAJOR	MINOR (from discipline) for SEM I, III, V, VII	MINOR (from vocational) for SEM II,IV,VI,VIII	MDC (other than discipline) for SEM I, II, III
ZOOLOGY	Physics	Awareness of law and Rights	Computer Application
	Chemistry	Computer Science & IT	Mathematics
	Botany	Mental Health	Statistics
	Geology	Organic Farming	Banking and Financial services
		Creative writing (Hindi)	Commerce
		Entrepreneurship skill	Management
		Medical Lab & Techniques	Ancient Indian history & Culture
			Anthropology
			Bengali
			Economics
			English
			Gandhian thoughts
			Geography
			Hindi
			History
			Home science
			Labour and Social Welfare
			Maithili
			Music
			Persian
			Philosophy
			Political science
			Psychology
			Public Administration
			Rural Economics
			Sanskrit
			Santhali
			Sociology
			Urdu

Note:

- Students will have to choose one Minor (from discipline) paper and have to study this Minor paper in respective four odd semester (SEM I, III, V, VII).
- Student will have to choose one MDC (other than discipline) paper in three semesters (SEM I, II, III). In the above-mentioned semester students have to change MDC subject. Students are not allowed to choose MDC subject which was studied in Intermediate.

Question format for 75 marks: (Major and Multidisciplinary)

Subject/code		Exam Year
F.M. =75	P.M. = 30	
Time = 3 Hrs.		

General Instructions:

- i. **Group A** carries very short answer type **compulsory** question.
- ii. **Answer 4 out of 6** subject/descriptive questions given in **Group B**.
- iii. Answer in your own words as far as practicable.
- iv. Answer all sub parts of a question at one place.
- v. Numbers in right indicate full marks of the question.

Group A

1. [5x1 = 5]

- i.
- ii.
- iii.
- iv.
- v.

2. [5]

3. [5]

Group B

4. [15]

5. [15]

6. [15]

7. [15]

8. [15]

9. [15]

Note: There may be subdivisions in each question asked in Theory Examination.

Subject/code		Exam Year
F.M. =60	P.M. = 24	
Time = 3 Hrs.		

General Instructions:

- i. **Group A** carries very short answer type **compulsory** question.
- ii. **Answer 3 out of 5** subject/descriptive questions given in **Group B**.
- iii. Answer in your own words as far as practicable.
- iv. Answer all sub parts of a question at one place.
- v. Numbers in right indicate full marks of the question.

Group A

1. [5x1 = 5]

- i.
- ii.
- iii.
- iv.
- v.

2. [5]

3. [5]

Group B

4. [15]

5. [15]

6. [15]

7. [15]

8. [15]

9. [15]

Note: There may be subdivisions in each question asked in Theory Examination.

PRACTICAL

MAJOR & MINOR PAPER: END SEMESTER EXAMINATION

MODEL OF QUESTIONS & DISTRIBUTION OF MARKS

SEMESTER: II (Major Paper- 03)

Full Marks : 75		Time : 03 hrs	
SN	Model of questions	Marks	
1	Dissection	15	
2	Mounting	05	
3	Spotting: Museum Specimens (02), Slides(02) , Bone(01) 5x5=	25	
4	Project/Collection/Model/Chart	15	
5	Record	10	
6	Viva	05	
Pass percentage 40% (30 Marks)		Total=	75

- Pass marks for internal (40%)=10 Marks

SEMESTER: III (Major Paper- 05)

Full Marks :75		Time : 03 hrs	
SN	Model of questions	Marks	
1	Economic Zoology 5x2=	10	
2	Evolution 5x2=	10	
3	Spotting: 3 Bones,1 Evolution & 1 Economy Zoology 5x5=	25	
4	Project/Collection/Model/Chart	15	
5	Record	10	
6	Viva	05	
Pass percentage 40% (30 Marks)		Total=	75

- Pass marks for internal (40%)=10 Marks

SEMESTER: IV (Major Paper- 08)

Full Marks :75		Time : 03 hrs	
SN	Model of questions	Marks	
1	Physiology	15	
2	Biochemistry	10	
3	Spotting: Endocrine & Reproductive Biology (5 Slides) 5x5=	25	
4	Tools and Techniques	10	
5	Record	10	
6	Viva	05	
Pass percentage 40% (30 Marks)		Total=	75

- Pass marks for internal (40%)=10 Marks

SEMESTER: V (Major Paper- 11)

Full Marks :75		Time : 03 hrs	
SN	Model of questions	Marks	
1	Cell division (Temporary Slide preparation (Cell division)- 1 Experiment	10	
2	Problems based on Mendelian ratios	15	
3	Spotting: 3 cell biology & 2 from genetics 5x5=	25	
4	Project/Collection/Model/Chart	15	
5	Record	10	
6	Viva	05	
Pass percentage 40% (30 Marks)		Total=	75

- Pass marks for internal (40%)=10 Marks

SEMESTER: VI (Major Paper- 15)

Full Marks :75		Time : 03 hrs	
SN	Model of questions	Marks	
1	Immunology	10	
2	Toxicology	10	
3	Spotting: 2 immunology, 2 Developmental Biology, 1 toxicology-pesticides/Herbicides 5x5=	25	
4	Project/Collection/Model/Chart	15	
5	Record	10	
6	Viva	05	
Pass percentage 40% (30 Marks)		Total=	75

- Pass marks for internal (40%)=10 Marks

SEMESTER: VII (Major Paper- 19)

Full Marks :75		Time : 03 hrs	
SN	Model of questions	Marks	
1	Microbiology	10	
2	Biotechnology	10	
3	Biostatistics	15	
4	Histology and Histochemistry	10	
5	Spotting: Microbiology 5x3=	15	
6	Record	10	
7	Viva	05	
Pass percentage 40% (30 Marks)		Total=	75

- Pass marks for internal (40%)=10 Marks

Minor Paper-01: END SEMESTER EXAMINATION

SEMESTER I Minor Paper-1 Practical		
Full Marks : 25		Time : 3hrs
SN	Model of questions	Marks
1	Evolution	07
2	Slide observation and Explanation	04
3	Spotting: (2 museum specimens, 2 permanent Slides) 4x2=	08
4	Record	03
5	Viva	03
	Total=	25
Pass Percentage: 40% (10 marks)		

Minor Paper-02: END SEMESTER EXAMINATION

SEMESTER III Minor Paper-2 Practical		
Full Marks : 25		Time : 3hrs
SN	Model of questions	Marks
1	Biochemistry	10
2	Model study 3x1=	03
3	Spotting: (2 Permanent Slides, 1Parental care) 2x3=	06
4	Record	03
5	Viva	03
	Total=	25
Pass Percentage: 40% (10 marks)		

Minor Paper-03: END SEMESTER EXAMINATION

SEMESTER V Minor Paper-3 Practical		
Full Marks : 25		Time : 3hrs
SN	Model of questions	Marks
1	Endocrine (Demo)	10
2	Economic Zoology 2x2=	04
3	Spotting: (3 Histology Endocrine glands & 1 embryology) 2x4=	08
4	Record	03
5	Viva	03
	Total=	25
Pass Percentage: 40% (10 marks)		

Minor Paper-04: END SEMESTER EXAMINATION

SEMESTER VII Minor Paper-04 Practical		
Full Marks : 25		Time : 3hrs
SN	Model of questions	Marks
1	Biochemical test	08
2	Preparation of blood film	05
3	Spotting: (Phenotypic traits of drosophila) 3x2=	06
4	Record	03
5	Viva	03
	Total=	25
Pass Percentage: 40% (10 marks)		

Advance Major Paper-03: END SEMESTER EXAMINATION

SEMESTER VIII Advance Major Paper-03 Practical		
Full Marks : 75		Time : 3hrs
SN	Model of questions	Marks
1	Blood based experiment	15
2	Biostatistics	10
3	Computer applications	10
4	Biochemistry experiment	10
5	Research Methodology	15
6	Record	10
7	Viva	05
	Total=	75
Pass Percentage: 40% (10 marks)		

Course objectives: The primary objective of the course in semester-I and semester-II is to make acquaintance with different life forms on earth and drive home the relationship between different living forms both at the genetic and the ecological level. It will provide an opportunity to have a novel branch of science dealing with identification and assigning exact position in animal kingdom.

NON-CHORDATES:

Unit –I

1. **General characters and classification up to class**
2. **Protozoa:**
 - 2.1 Locomotion, Nutrition & Reproduction (General)
 - 2.2 Life cycle and pathogenicity of Leishmania
3. **Porifera:**
 - 3.1 Canal system (General)
 - 3.2 Life cycle of sycon

Unit –II

1. **Cnidaria:**
 - 1.1 Life cycle of Obelia
 - 1.2 Coral reefs
2. **Ctenophora:** Affinities
3. **Platyhelminthes:** Parasitic adaptation (General)

Unit –III

1. **Nemathelminthes:**
 - 1.1 Life cycle of *Ascaris lumbricoids*
 - 1.2 Life cycle of *Wuchereria bancrofti*
2. **Annelida:**
 - 2.1 Metamerism
 - 2.2 Type study: Leech and Earthworm (Digestive, Excretory, Circulator system – Comparative)
3. **Arthropoda:**
 - 3.1 Larval forms of crustacean
 - 3.2 Mouth parts in insects
 - 3.3 Vision (Structure of eye)in prawn and cockroach

Unit –IV

1. **Mollusca:**
 - 1.1 Torsion and detorsion in gastropod
 - 1.2 Respiration in Mollusca (Unio & Pila)
2. **Echinodermata:**

- 2.1 Water vascular system in Holothuria
- 2.2 Larval forms of Echinodermata
- 3. **Hemichordates:** General characters and affinities

Recommended books

Non-chordates

1. Barnes, R.D. Invertebrate Zoology-(W.B. Saunders Co.)
2. Invertebrate structure and function: Barrington (Nelson)
3. Kotpal, Agarwal & Khetrapal: Modern Textbook of zoology: Invertebrate (Rastogi publication)
4. R.L. Kotpal: Invertebrate series – Protozoa to Minor phyla: (Rastogi publication)
5. Practical zoology INVERTEBRATE- S.S. LAL
6. Nigam: Biology of non-chordates (1997, S Chand)

CHORDATES:

Unit –I

1. **Chordates:**
 - 1.1 General characters and classification up to orders
2. **Urochordates & Cephalochordates:**
 - 2.1 General characters with example
 - 2.2 Retrogressive metamorphosis in Herdmania
3. **Cyclostomes:**
 - 3.1 General characters

Unit –II

1. **Pisces:**
 - 1.1 Type study- Labeo and Scoliodon (General anatomy & Circulation)
 - 1.2 Migration in Fishes
 - 1.3 Dipnoi
2. **Amphibia:**
 - 2.1 Parental care in Amphibia
 - 2.2 Neoteny

Unit –III

1. **Reptiles:**
 - 1.1 Study of Skull
 - 1.2 Biting mechanism of snakes
2. **Aves:**
 - 2.1 Flight adaption
 - 2.2 Migration in birds
3. **Mammals:**
 - 3.1 Prototheria
 - 3.2 Metatheria

Unit –IV

1. **Comparative anatomy of vertebrates:**
 - 1.1 Skin
 - 1.2 Heart
 - 1.3 Aortic Arches
 - 1.4 Brain

Questions to be set from both units:

NON CHORDATES AND CHORDATES

Unit –I NON CHORDATES:

1. STUDY OF PERMANENT SLIDES:
 - 1.1 Protozoa: *Amoeba*, *Entamoeba*
 - 1.2 Porifera :T.S & L.S. of *Sycon*
 - 1.3 Coelenterata: *Obelia* Colony
 - 1.4 Platyhelminthes: *Fsaciola* (wm)
2. STUDY OF MUSEUM SPECIMENS:
 - 2.1 Porifera: *Sycon*
 - 2.2 Coelenterata: *Aurelia*, *Physalia*
 - 2.3 Helminthes: *Fasciola hepatica*, *Ascaris*

Unit –II CHORDATES:

1. DISSECTION:
 - 1.1 Scoliodon: General anatomy
 - 1.2 Afferent and Efferent blood vessels
2. MOUNTING:
 - 2.1 Placoid, Cycloid, Ctenoid scales of fishes
3. STUDY OF PERMANENT HISTOLOGICAL SLIDES:
 - 3.1 Mammal: Testes, Ovary, Skin, Liver, Pancreas, Stomach, Intestine
4. STUDY OF MUSEUM SPECIMENS:
 - 4.1 Fish: Hammer headed shark, *Hippocampus*, *Anabas testudeneus*, *Catla catla*, *Labeo rohita*, *Amphibia*, *Ichthyophis*, *Hyla*,
5. OSTEOLOGY:
 - 5.1 Amphibia, and Mammal: Vertebrae, Limb bones and Girdle bones
6. SPECIMEN COLLECTION
7. CHART AND MODEL PREPARATION (Comparative Anatomy)

Course Learning Outcomes: Having completed Semester-I and Semester-II, the students would develop an aptitude for classical Zoology which is integral part for understanding the life forms with a vision of comparative account of different systems of the animals. Without having wide spectrum knowledge of life forms, the concept of biodiversity will be hard to understand and especially, their identification. This paper will enable the students to create interest in animal world who will carry forward their knowledge to the future generation to unravel many mysteries of animal forms.

Recommended books

Chordates

1. RL Kotpal: Modern textbook of zoology VERTEBRATE
2. Young, J.Z: Life of Vertebrates (Oxford University Press)
3. Hildebrand: Analysis of vertebrates Structure (Wiley)
4. Practical zoology VERTEBRATE- S.S. LAL
5. Nigam: Biology of Chordates (1997, S Chand)
6. Parker & Haswell: Text Book of Zoology, Vol. II (2005, Macmillan)

Course Objectives: This paper contains multitude of sections throwing light upon some essential and basic aspects of the subject. The skeletal structure dealing with various limb bones vertebrae and girdles in different vertebrate groups pertains to throw light on comparative account of subtle differences present in them.

This paper aims at giving focus on the theories of evolution for giving comprehensive idea origin of various life forms on the earth. This paper unravels the mystery of origin life and their subsequent evolution giving spectrum of life forms.

Not only this, the paper has been framed to provide opportunity to the students to develop skill development and employment generation by adopting small scale industries of animal products such as silk, honey, pearl, fish, lac etc. of great economy value.

Additionally the concept of animal behaviour in different animals which are generally different from human behaviour have been incorporated to give students a comprehensive idea a few interacting phenomena such as bird migration, parental care in fish and amphibian in this paper.

OSTEOLOGY, EVOLUTION, ECONOMIC ZOOLOGY AND ANIMAL BEHAVIOUR

Unit –I

OSTEOLOGY (Fish, Amphibia, Reptilia, Aves, Mammal) Skull Bones

- 1.1 Limb Bones
- 1.2 Girdle Bones
- 1.3 Vertebrae

Unit –II

EVOLUTION

- 1.1 Origin and evolution of life on earth
- 1.2 Theories of Evolution - Darwinism, Neo Darwinism, Lamarckism, Neo Lamarckism, Synthetic theory of Evolution
- 1.3 Concept of isolating mechanism and its role in evolution

Unit-III

ECONOMIC ZOOLOGY

- 1.1 Sericulture
- 1.2 Apiculture
- 1.3 Lac culture
- 1.4 Pearl culture
- 1.5 Pisciculture

Unit-IV

ANIMAL BEHAVIOUR

- 4.1 Innate and Learned Behaviour
- 4.2 Social Behavior in Insects (Honey Bee, Termite)
- 4.3 Parental Care (Fishes and Amphibia)
- 4.4 Nesting and Brooding behaviour in Birds
- 4.5 Migration and Navigation in Birds
- 4.6 Schooling in Fishes

BONES, EVOLUTION, ECONOMIC ZOOLOGY AND ANIMAL BEHAVIOUR**Unit-I****BONES (Fish, Amphibia, Reptilia, Aves, Mammals)**

- 1.1 Skull Bones
- 1.2 Limb Bones
- 1.3 Girdle Bones
- 1.4 Vertebrae

Unit-II**EVOLUTION**

- 2.1 Homology and Analogy in wings (Butterfly, Bird, and Bat)
- 2.2 Serial Homology in appendages of Prawn
- 2.3 Adaptive Radiation in Beaks of Birds
- 2.4 Adaptive Radiation in Feet of Birds
- 2.5 Adaptive Radiation in Dentition of Mammal

Unit-III**ECONOMIC ZOOLOGY****Economic importance of the following:**

- 3.1 Silk worm
- 3.2 Honey Bee
- 3.3 Lac Insect
- 3.4 Fishes – *Labeo rohita*, *Catla catla*, *Clarius batracus*, *Heteropneustes fossilis*, *Channa punctatus*.

Unit-IV**ANIMAL BEHAVIOUR**

- 4.1 Specimen showing Parental care in Fish and Amphibia
- 4.2 Comment upon models of Animal Behaviour

Course Outcomes: The students will find a new inclusion in terms of osteology in theory paper as it had been generally part of practical papers. This will give opportunity to them to study details of various bones in different vertebrates. Besides, the concept of evolution will unravel persisting question in minds of students by this brief but highly effective topics. The economic zoology will give opportunity to the students for skill development and employment generation. The concept of animal behavior and a few manifestations will help them to understand it in scientific manner rather than social context.

Recommended books

Osteology

1. Introduction to Human Osteology (publisher: [Grand Valley State University](#))
2. Human Microscopic Anatomy: R.V. Krstic, Springer - Verlay
3. RL Kotpal: Modern textbook of zoology VERTEBRATE

Evolution

1. Strik Berger, M.W. Evolution, Jones & Bartett. Publishers, Boston, London
2. An Introduction to Paleontology –A.P.Tyagi(S.Chand & Com.LTD)
3. Hall and Hallgrimsson: Strickberger's Evolution (2008, Jones and Bartlett)
4. Moody: Introduction to Evolution (1978, Kalyani).
5. Rastogi: Organic Evolution (2007, Kedarnath & Ramnath)
6. Futuyma: Evolutionary Biology (2005, Sinauer)

Economic Zoology

1. Shukla and Upadhyaya: Economic Zoology (Rastogi Publishers, 1999-2000)
2. Shrivastava: Test book of Applied Entomology, Vol. I &II (Kalyani Publishers, 1991)
3. Mani: Insects, NBT, India, 2006.
4. Jabde: Text Book of Applied Zoology: Vermiculture, Apiculture, Sericulture, Lac culture,
5. Agricultural Pests and their Control, 2005 Publisher Vedams eBooks (P) Ltd. New Delhi

Animal behaviour

1. An introduction to animal behavior: Aubrey Manning and Dawkins
2. Animal Behavior: Reena Mathur, (Rastogi Publication)
3. Animal behavior: VK Agarwal
4. Animal behavior: Natrajan and Arumugam
5. Alcock: Animal Behaviour: An evolutionary approach (9 ed. 2009, Sinauer)
6. Drickamer, Vessey and Jacob: Animal Behaviour (5th ed. 2002, McGraw Hill)
7. Goodenough et al.: Perspectives on Animal Behaviour (1993, Wiley)
8. Grier: Biology of Animal Behaviour (1984, Mosby)
9. Lorenz: The foundation of ethology (1981, Springer)
10. Manning & Dawkins: An Introduction to Animal Behaviour (5th ed. 1998, Cambridge).
11. Mcfarland: Animal Behaviour, Psychology, Ethology and Evolution (1985, Pitman).
12. Scott: Essential Animal Behaviour (2005, Blackwell)
13. Slater: An introduction to ethology (1985, Cambridge)

Course Objective: This paper gives a combination of interrelated sections which would help the students to imbibe the contents in totality. The basic inputs of physiology, endocrinology and reproductive biology have been placed to build a strong foundation for higher studies. The concept of hormones and their roles in reproduction have been considered together to have a composite idea of the entire phenomena.

PHYSIOLOGY, ENDOCRINOLOGY AND REPRODUCTIVE BIOLOGY

Unit-I

PHYSIOLOGY

- 1.1 Digestion (GIT and associated glands) and Absorption (Carbohydrate, Protein and Lipid)
- 1.2 Respiration (Anatomy of respiratory system, Mechanism and Control of Breathing, Transport of Gases)
- 1.3 Circulation (Anatomy of Mammalian heart, Cardiac cycle and ECG, Composition of Blood and Lymph, Blood Clotting mechanism, Blood Group – ABO)
- 1.4 Nervous System (Structure of Neuron, Nerve conduction, Reflex action)
- 1.5 Excretion (Structure and function of Kidney, Mechanism of Urine formation)

Unit-II

ENDOCRINOLOGY AND REPRODUCTIVE BIOLOGY

- 2.1 Structure and function of Endocrine glands and related disorders :
 - 2.1.1 Pituitary
 - 2.1.2 Pineal
 - 2.1.3 Thyroid
 - 2.1.4 Adrenal
 - 2.1.5 Gonads
 - 2.1.6 Islets of Langerhans
- 2.2 Mechanism of Hormone action (Peptide and Steroid Hormone)
- 2.3 Reproductive cycle (Estrous cycle and Menstrual cycle)

Course Outcomes: The students will consolidate the concept of physiology, endocrinology and reproductive biology within a short span of time. The contents are scrupulously placed to inculcate the habit of studying the things in totality. The wide range of hormones has been incorporated to make coherence with respect to interplay of hormones in metabolic pathways. Especially the hormonal disorder in human population will further initiate inquisitive behavior in the students to manage and serve the people around them.

Course Objective: The concept of biomolecules is fundamental in understanding almost all sections of Zoology and especially physiology, cytology and molecular biology are based upon the concept of biochemistry. Its early inclusion in syllabus aims at providing essential ingredients required to understand all sections with a strong foundation of biochemistry.

The tools and techniques consisting of various instruments will prepare the students in advance for developing research aptitude in the students. Their applications in pathological and biochemical tests are of immense significance.

BIOCHEMISTRY, TOOLS AND TECHNIQUES

Unit-I

BIOCHEMISTRY

- 1.1 Structure , Classification and Significance
 - 1.1.1 Carbohydrate
 - 1.1.2 Protein
 - 1.1.3 Lipid
- 1.2 Metabolic Pathways
 - 1.2.1 Glycolysis
 - 1.2.2 Glycogenesis
 - 1.2.3 Gluconeogenesis
 - 1.2.4 Krebs Cycle
- 1.3 Enzymes: Classification, nature and mechanism of action
- 1.4 Vitamins: Classification, sources, function and deficiency

Unit-II

TOOLS AND TECHNIQUES

- 2.1 Working principle, parts, application of instruments
 - 2.1.1 pH meter
 - 2.1.2 Centrifuge
 - 2.1.3 Colorimeter
 - 2.1.4 Spectrophotometer
- 2.2 Microscopy: Working principle, parts, application
 - 2.1.1 Bright field
 - 2.1.2 Dark field
 - 2.1.3 Phase contrast
 - 2.1.4 Fluorescence

Course Outcomes: The topics of biochemistry and tools & techniques will increase the horizon of knowledge in the students as they will have new exposure of this field which will be of utmost importance in developing scientific temper in them. Especially quantitative estimation of biochemical parameters as application the instruments will be quite handy in dealing with research assignment.

PHYSIOLOGY, ENDOCRINOLOGY AND REPRODUCTIVE BIOLOGY, BIOCHEMISTRY, TOOLS AND TECHNIQUES

Unit-I

PHYSIOLOGY

- 1.1 Estimation of Hb %
- 1.2 Determination of blood group
- 1.3 Estimation of blood glucose
- 1.4 Bleeding and Clotting time
- 1.5 RBC and WBC counting

UNIT-II

ENDOCRINOLOGY AND REPRODUCTIVE BIOLOGY

- 2.1 Histological Slides: Pituitary, Pineal, Thyroid, Adrenal, Gonads, Islets of Langerhans
- 2.2 Stages of Estrous cycle

Unit-III

BIOCHEMISTRY

- 3.3 Test for
 - 3.3.1 Carbohydrate – Iodine test, Benedict's test
 - 3.3.2 Protein - Biuret test, Millon's test
 - 3.3.3 Lipid – Solubility test, Acrolein test
 - 3.3.4 Test for Salivary amylase enzyme activity and importance of temperature and pH on salivary amylase action

Unit-IV

TOOLS AND TECHNIQUES

- 4.1 Working principle, parts, application of instruments
 - 4.1.1 pH meter
 - 4.1.2 Centrifuge
 - 4.1.3 Colorimeter
 - 4.1.4 Spectrophotometer
- 4.2 Microscopy: Working principle, parts And its application
 - 4.2.1 Bright field
 - 4.2.2 Dark field
 - 4.2.3 Phase contrast
 - 4.2.4 Fluorescence

Recommended books

Physiology

1. Review of Medical physiology, Ganong (Lange)
2. Animal Physiology - Neilsen (Cambridge)
3. Animal Physiology - Eckert, R. (W. H. Freeman)
4. Review of Medical physiology, Ganong (Lange)
5. Reproductive Physiology - (Nalbandov, A.V.)
6. General & Comparative Physiology - Hoar (Prentics Hall)
7. Animal Physiology - Neilsen (Cambridge)
8. Comparative Animal Physiology - Progser (Satish Book Enterprise)

Endocrinology and reproductive biology

1. Endocrinology - Hadley.
2. General Endocrinology - Bagnara, and Turner (W.B. Saunders)
3. Endocrinology - Hadley'(Prentice Hall)
4. E.J.W. Barrington - General & Comparative Endocrinology, Oxford, Clarendon Press.
5. P.J.Bentley, Comparative Vertebrate Endocrinology, Cambridge University Press.
6. R.H. Williams - Text Book of Endocrinology. W.B. Saunders.
7. C.R. Martin - Endocrine Physiology, oxford.
8. A Gorbman et al. Comparative Endrocrinology. John Willey & Sons

Biochemistry

1. Cox, M.M and Nelson, D.L. (2008). Lehninger Principles of Biochemistry, V Edition, W.H. Freeman and Co., New York.
2. Berg, J.M., Tymoczko, J.L. and Stryer, L. (2007). Biochemistry, VI Ed, W.H. Freeman and Co., New York.
3. Murray, R.K., Bender, D.A., Botham, K.M., Kennelly, P.J., Rodwell, V.W. and Well, P.A. (2009), Harper's Biochemistry (25th ed. 2000, Appleton & Lange)
4. Illustrated Biochemistry, XXVIII Edition, International Edition, The McGraw-Hill Companies Inc.
5. Hames, B.D. and Hooper, N.M. (2000). Instant Notes in Biochemistry, II Edition, BIOS Scientific Publishers Ltd., U.K.
6. Boyer: Concepts in Biochemistry (3rd ed. 2006, Brooks/Cole)
7. Conn, Stumpf, Bruening & Doi: Principles of Biochemistry (5th ed. 1987, Wiley)

Tools and techniques

1. Roymahoney: Laboratory Techniques in Zoology (1966, Butterworths).
2. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,)
3. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill- Livingstones)
4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill)
5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)

Course Objectives: The fundamentals of cell biology have been prescribed to understand basic structure of components of cells. The functional aspects cell constituents will help the students to cell physiology as an adjunct to molecular biology. The latter will provide the details DNA and functional aspect of genes.

CELL BIOLOGY AND MOLECULAR BIOLOGY

Unit-I

CELL BIOLOGY

- 1.1 Prokaryotic and eukaryotic cell
- 1.2 Plasma membrane (models of plasma membrane structure; Transport across membranes: active, passive and facilitated transport; Cell junctions: Occluding junctions (Tight junctions), anchoring junctions (desmosomes) and communicating junctions (gap junctions).
- 1.3 Endomembrane system (structure and function of endoplasmic reticulum, Golgi complex, and lysosome)
- 1.4 Cytoskeleton (Structure and functions of microtubules, microfilaments and intermediate filaments)
- 1.5 Structure and functions of mitochondria
- 1.6 Nucleus (Ultra-structure of nucleus, Nuclear Envelope, Nuclear pore complex and Nucleolus; Chromatin: Euchromatin and Heterochromatin, packaging of DNA: nucleosome).
- 1.7 Cell division and its significance (Mitosis, meiosis and its regulation)
- 1.8 Polytene and Lampbrush chromosomes

Unit-II

MOLECULAR BIOLOGY

- 2.1 DNA structure and types
- 2.2 DNA Replication (*E. coli* and eukaryote)
- 2.3 DNA damage and repair mechanism
- 2.4 RNA structure and types
- 2.5 Genetic code
- 2.6 Protein synthesis (transcription and translation)
- 2.7 Regulation of gene expression (*Lac* operon, *Trp* operon)

Course Outcomes: The cell and molecular biology taught after going through biochemistry will certainly pave the way for developing better understanding of functional aspect of cell at molecular level. Gene regulation and protein synthesis arising out of it will have significant impact in exploring novel areas of research in molecular biology.

Course Objectives: The classical genetics which incorporates chromosomal level inheritance with concept of genes and alleles. It explains how Mendel's particulate mechanism differed from blending theory of inheritance. The terms: monohybrid cross, dihybrid crosses, dominant, recessive, extra-chromosomal inheritance are very basic to begin studying genetics. The contents incorporated will explain how variations arose in the nature.

Unit-I**GENETICS**

- 1.1 Mendel's law of inheritance
- 1.2 Monohybrid cross and law of segregation
- 1.3 Dihybrid cross and law of independent assortment
- 1.4 Extension of Mendelism (Incomplete dominance and co-dominance, Multiple alleles, Lethal alleles, Epistasis, Pleiotropy, Epistasis, Penetrance and Expressivity, Sex-linked inheritance, sex-influenced inheritance and sex-limited inheritance)
- 1.5 Cytoplasmic inheritance (Criteria for extra-chromosomal inheritance, Mitochondrial mutations in *Saccharomyces*, Infective heredity in *Paramecium* and Maternal effects in snail)
- 1.6 Linkage and crossing-over
- 1.7 Structural and numerical alterations of chromosomes
- 1.8 Gene mutation
- 1.9 Environmental and Chromosomal mechanisms of sex determination in animals (*Drosophila*, reptiles, hen, Man)

Course Outcomes: Classical genetics provide many insights into inherited traits and elucidate many inherited human disorders that are known to follow Mendel's law of inheritance and are useful to explain the appearance of diseases within families. The actual process of gene mapping and mutant gene identification can be ascertained by having knowledge of contents of this paper. It is the foundation for all other areas in genetics.

CELL BIOLOGY AND MOLECULAR BIOLOGY, GENETICS

Unit-I

CELL BIOLOGY AND MOLECULAR BIOLOGY

- 1.1 Drawing of ultrastructure of cell and different organelles (from photographs provided)
- 1.2 Study of various stages of mitosis from permanent slides.
- 1.3 Study of various stages of meiosis from permanent slides.
- 1.4 Preparation of temporary stained squash of onion root tip to study various stages of mitosis.
- 1.5 Preparation of polytene chromosome from *Chironomus* larva

Unit-II

GENETICS

- 1.1 Study of phenotypic traits in *Drosophila*
- 1.2 Demonstration of Mendelian monohybrid cross and dihybrid cross in *Drosophila*
- 1.3 Demonstration of sex-linked inheritance in *Drosophila* making a cross between white eye and wild type flies
- 1.4 Demonstration of human karyotypes and numerical alterations (Down syndrome, Klinefelter syndrome and Turner syndrome).

Recommended books

Cell biology and molecular biology

1. Karp, G. (2010). Cell and Molecular Biology: Concepts and Experiments. VI Edition. John Wiley and Sons. Inc.
2. De Robertis, E.D.P. and De Robertis, E.M.F. (2006). Cell and Molecular Biology. VIII Edition. Lippincott Williams and Wilkins, Philadelphia.
3. Cooper, G.M. and Hausman, R.E. (2009). The Cell: A Molecular Approach. V Edition. ASM Press and Sunderland, Washington, D.C.; Sinauer Associates, MA.
4. Becker, W.M., Kleinsmith, L.J., Hardin, J. and Bertoni, G. P. (2009). The World of the Cell. VII Edition. Pearson Benjamin Cummings Publishing, San Francisco.
5. Bruce Albert, Bray Dennis, Lewis Julian, Raff Martin, Roberts Keith and Watson James (2008). Molecular Biology of the Cell, V Edition, Garland publishing Inc., New York and London
6. Molecular Biology of the Gene - Watson, J.D et al (Benjamin/ Cummings)
7. De Robertis and De Robertis: Cell and Molecular Biology (Sauders College)
8. Alberts et al: Molecular Biology of the Cell (2008, Garland)
9. Lodish et al: Molecular Cell Biology (2008, Freeman)
10. Cooper and Hausman: The Cell A Molecular approach (2007, Sinauer)
11. Lininger Principles of biochemistry: Cox & Nelson, MacMillan & Freeman, USA

Genetics

1. Molecular biology of Gene: Watson et al., Pearson Publication, USA
2. Strickberger's Genetics, Prinitis Hall of India (PHI), Delhi
3. Principles of Genetics: Snustad& Simmons, John Wiley & Sons, USA
4. Modern Genetics Analysis: Integrating Genes and Genomes, Griffith et al., W. H. Freeman & Company, USA
5. Genetics: Russell & Benjamin, Cummings Publishing Company, USA.
6. Genetics: PK Gupta, Rastogi Publication, New Delhi.
7. Gene regulation: Latchmann, Taylor & Francis, USA
8. Genetics: Benjamin A pierce

Course Objectives: The lecture-discussion part is conceptualized with the aim that students are taught the basics of immunology so as to develop understanding of the subject such as “how does the immune system work”? The molecular and cellular components and pathways that protect an organism from various antigenic agents have been dealt in this section.

The related section of parasitology has been incorporated to find a few parasites causing some common diseases in human beings of which some are very harmful to cause severe health conditions. How host and parasites are showing specificity by their biological rhythm has been the highlight of this paper.

IMMUNOLOGY AND PARASITOLOGY

IMMUNOLOGY

Unit-I

- 1.1 Overview of Immune System
- 1.2 Cells and organs of the Immune system
- 1.3 Innate and Adaptive Immunity (Anatomical barriers, Inflammation, Cell and molecules involved in Innate immunity, Adaptive immunity: Cell mediated and humoral, Passive: Artificial and natural immunity, Active: Artificial and natural Immunity)

Unit-II

- 2.1 Antigens
- 2.2 Immunoglobulins (Structure and functions of different classes of immunoglobulins, Antigen-antibody interactions)

Unit-III

- 3.1 Immunoassays (ELISA and RIA).
- 3.2 Vaccines

PARASITOLOGY

Unit-I

- 1.1 General concept of parasites and parasitism: Introduction to parasitology: basic principles and concepts
- 1.2 Signs and symptoms of parasitic diseases
- 1.3 Vector, Zoonosis

Unit-II

- 2.1 Parasitic protozoans: Life cycle, Pathogenesis and Disease caused by: *Amoebae*, *Plasmodium*, [*Trypanosoma*](#)
- 2.2 Parasitic Helminths: Life cycle, Pathogenesis and Disease caused by- *Trichinella spiralis*, *Echinococcus granulosus*

Unit-III

- 1.1 Dynamics of Host-Parasite interaction
- 1.2 Parasitic adaptations

Course Outcomes: Upon completion of the course students have a sound understanding of the essential elements of the immune system, preparing them to engage further in this rapidly evolving field. The students will be able to identify the cellular and molecular basis of immune responsiveness and understand how the innate and adaptive immune responses coordinate to fight invading pathogens. It will encourage the students to explore designing new vaccines and other immunotherapeutic strategies. The study of parasites, their life cycles, pathogenocity, prophylaxis and prevention of outbreak epidemics will be a boon for the students to tide over adverse situations.

Course Objectives: The detail process of embryonic development up to implantation will provide a comprehensive idea of developmental biology. The topics have been scrupulously placed to have systematic approach of embryology of different vertebrate groups

DEVELOPMENTAL BIOLOGY

Unit-I

- 1.1 Gametogenesis: Spermatogenesis, Oogenesis
- 1.2 Types of eggs, Cleavage patterns and Significance
- 1.3 Gastrulation: Invagination, Involution, Ingression, Delamination, Epiboly

Unit-II

- 2.1 Concept of competence, determination and differentiation of cell
- 2.2 Early development of frog and chick up to gastrulation

Unit-III

- 3.1 Extra embryonic membranes
- 3.2 Fate of Germ Layers; Implantation of embryo in humans
- 3.3 Placenta: Structure, Types and Function
- 3.4 Metamorphosis in frog and insect

Course Outcomes: The students will come to know the process by which a fertilized egg is transformed into a multicellular organism. It explains the fundamental concept of embryogenesis and organogenesis as well.

Course objectives: The objective of this course to make awareness among the students about the surrounding environment, common ecological phenomena such as pollution, population, community with emphasis on wildlife conservation has been underscored.

ECOLOGY AND TOXICOLOGY

Unit-I

ECOLOGY

- 1.1 Concept of ecology
- 1.2 Biosphere: Lithosphere, Hydrosphere, Atmosphere
- 1.3 Ecosystem: Types of ecosystems, Food chain, Food web, Ecological pyramids, Energy flow through the ecosystem, Nutrient and Biogeochemical cycle
- 1.4 Population: Population Density, Natality, Mortality, Life tables, Fecundity tables, Survivorship curves, Age ratio, Sex ratio, Dispersal and Dispersion; Types of interaction between two species
- 1.5 Community: Species richness, Dominance, Biodiversity, Abundance, Vertical stratification; Ecotone and Edge effect; Ecological succession with one example
- 1.6 Pollution: Water Pollution, Soil Pollution and Air Pollution
- 1.7 Wildlife conservation

Unit-II

TOXICOLOGY

- 2.3 Basic concepts of Toxicology
- 2.4 Toxicological Dose Descriptors: LC_{50} , LD_{50} , EC_{50} , DT_{50}
- 2.5 Toxicants: Chemicals, Pesticides, Herbicides, Fertilizers, Heavy metals, Food additives, Radioactive substances

Course outcomes: Students will be exposed to the fundamental aspects of ecology. They will get idea about the impact of anthropogenic activities on the environment. Above all once they finish their course they are supposed become crusaders of sustainability of the environment and its conservation. They would feel connected with the environment.

The eco-toxicology will provide the students to find methodology to test the toxicity of various pollutants such as pesticides & heavy metals which are being integrated in food chains causing health hazards to the human population.

IMMUNOLOGY, DEVELOPMENTAL BIOLOGY AND TOXICOLOGY**Unit-I****IMMUNOLOGY**

- 1.1 Demonstration of lymphoid organs.
- 1.2 Histological study of spleen, thymus and lymph nodes through slides/ photographs
- 1.3 Preparation of stained blood film to study various types of blood cells.
- 1.4 ABO blood group determination.
- 1.5 Principle of different types of ELISA

Unit-II**DEVELOPMENTAL BIOLOGY**

- 1.1 Study of whole mounts of developmental stages of chick through permanent slides: Primitive streak (13 and 18 hours), 21, 24, 28, 33, 36, 48, 72, and 96 hours of incubation (Hamilton and Hamburger stages)
- 1.2 Study of different types of placenta (photomicrograph/ slides)

Unit-III**TOXICOLOGY**

- 1.1 Study of surfacing time and opercular beating of *Clarius batracus* under the different doses of DDT
- 1.2 Study of surfacing time and opercular beating of *Clarius batracus* under the two different toxicants

Recommended books**Immunology**

1. Kuby - Immunology W.H.Freeman, USA.
2. BIOS Instant Notes in Immunology
3. Kindit, T.J., Golds by R.A., Osborne, B.A. and Kuby, J (2006). Immunology, VI Edition. W.H.Freeman and Company.
4. David, M., Jonathan, B., David, R.B. and Ivan R. (2006). Immunology, VII Edition, Mosby, Elsevier Publication.
5. Abbas, K. Abul and Lechtman H. Andrew (2003) Cellular and Molecular Immunology. V edition. Saunders Publication.

Development biology

1. Introduction to Embryology - Balinsky (CBS College publishers)
2. Developmental Biology - Subramaniyan, T (Narosa publishing House)
3. Developmental Biology - Gilbert
4. Chordate Embryology: Verma & Agarwal: S.Chand & Com.LTD
5. BIOS instant notes Developmental biology

6. Wolpert: Principles of Development (3rd ed. 2007, Oxford)

Ecology

1. Odum and Barrett, Thomson, Ed. Brooks/Cole, Fundamentals of Ecology, Cengage Learning
2. Ecology, Environment and Resources Conservation: Singh, Singh and Gupta Ed., Anamaya Pub., New Delhi
3. Odum, E.P., Basic Ecology, Ed. Saunders College Pub.
4. Fundamentals of Ecology - Odum - (Saunders)
5. Krebs. C.J. Ecology - Harper & Row-N.York.
6. BIOS Instant Notes ECOLOGY
7. Concept of ecology- Kormondy
8. Textbook of environmental studies-Bharucha
9. Fundamentals of Environmental Biology-B. Mukherjee
10. Fundamentals of ecology- MC Das

Toxicology and Parasitology

1. KD Chatterjee: Parasitology
2. Burton J Bogitsh Human Parasitology 3rd edition Elsevier.
3. Roberts, L. S. and J. Janovy, Jr. 2004. Foundations of Parasitology. 7th Edition. McGraw Hill, Boston.
4. Smith. Animal Parasitology 1996. Cambridge University Press.
5. Chandler and Reid. Introduction to Parasitology 1970, Wiley.
6. Marr et al. Molecular Medical Parasitology 2003, Elsevier.
7. Lawrence R. Ash and Thomas C. Orihel. Atlas of Human Parasitology. American Society for clinical pathology press 5th edition, 2007.
8. Alan Gunn. Parasitology: An Integrated Approach Wiley-Blackwell. 2012.
9. Janet Amundson Romich. Understanding Zoonotic Diseases. 2007
10. Paul Schmid-Hempel. The Integrated Study of Infections, Immunology, Ecology, and Genetics (Oxford Biology), 2011
11. Alan Gunn. Parasitology: An Integrated Approach Wiley-Blackwell. 2012.
12. Janet Amundson Romich. Understanding Zoonotic Diseases. 2007
13. Paul Schmid-Hempel. The Integrated Study of Infections, Immunology, Ecology, and Genetics (Oxford Biology), 2011

Course Objectives: Microbiology has been proved to be one of the most significant in biology making it possible to define how some microorganisms cause diseases, discover treatments for such diseases and even use a few microbes for industrial applications. It provides deep understanding microorganisms: virus, bacteria, fungi and their impact on human health and environment. It also throws light on intricate relationships between microorganisms and larger animals. It will also guide towards safety and management of various diseases caused by them.

MICROBIOLOGY

Unit-I

- 1.1 Microbes: History of Microbiology, Naming, Classification and Identification of Microorganisms
- 1.2 Bacteria: Forms of Bacteria; External Structures: The wall and Membrane(s); Bacterial Internal Structures, Classification Systems for Bacteria
- 1.3 Viruses: viruses types

Unit-II

- 2.1 Microbial Diseases of the Upper Respiratory tract (Streptococcal Pharyngitis, the Common Cold), Diseases of the Lower Respiratory Tract (Tuberculosis, Pneumonia)- .
- 2.2 Normal Biota of the Gastrointestinal Tract, Bacterial Diseases of the Mouth (Oral Cavity) Dental Caries, Bacterial Diseases: Shigellosis (Bacillary Dysentery), Salmonellosis and Typhoid Fever, Cholera, Helicobacter
- 2.3 Urinary Tract Infections (UTI), Sexually Transmitted Diseases (STD) (Gonorrhea, Syphilis)
Mode of infection, diagnostic features, damages done, prophylaxis and cure.

Unit-III

- 3.1 Skin Diseases Caused by Microorganisms (Measles, Small Pox)
- 2.1 Cardiovascular and Lymphatic System Diseases Caused by Microorganisms (Septicemia, Sepsis and Septic Shock)-
Mode of infection, diagnostic features, damages done, prophylaxis and cure.

Course Outcomes: This paper will provide information to the undergraduate students to know various microorganisms causing particular diseases. It explains the importance of microorganisms in diagnosis, monitoring and treatment of infectious diseases. This will help in developing skill in the students to master hygiene procedure in connection with laboratory work and routine affairs of life. Especially such skilled students will prove their worth in adverse circumstances arising out of epidemics and pandemics.

Course Objectives: Major objective of this core paper is to introduce to the students contemporary molecular techniques for manipulation of genome that could assist them towards advanced understanding of biological processes in broad range of host organisms. The student should be able to understand standard and system-specific gene manipulation approaches ranging from bacteria to mammals. Protection of research outcomes has been given place in this paper.

BIOTECHNOLOGY

Unit-I

- 1.1 Principles of recombinant DNA technology – Restriction endonucleases, polymerases, ligase
- 1.2 Gene transfer: Transformation, Episomes, Plasmid, other cloning vector

Unit-II

- 2.1 PCR and RT-PCR
- 2.2 Biotechnological tools and techniques: Basic principles of electrophoresis
- 2.3 Blotting (Northern and Southern hybridization)
- 2.4 DNA fingerprinting and its uses
- 2.5 Gene sequencing: Shotgun method

Unit-III

- 3.1 Genetically modified animals and their importance
- 3.2 Therapeutic products produced by genetic engineering- blood proteins, human hormones, vaccines
- 3.3 Genomics, proteomics

Unit-IV

- 4.1 Indian patent law
- 4.2 Intellectual property rights and its legal protection in research

Course Outcomes: On successful completion of the course the students will possess the technical background knowledge needed to support biotechnology research activity. The students will show their understanding to file patent and IPR. This will encourage the students to use their skill in the field of biomedicine and biotechnology based industries which will fetch foreign exchange and increase GDP of country.

Course objectives: This paper deals with biostatistics with an idea to infuse significance of statistics in biological science, especially while conducting research to find a conclusion. The application of biostatistics in assessing problems related to human health is very beneficial.

The study of histology on the other hand will demonstrate an understanding and visualization of relationships of cells, tissues and organs of human body.

The objective of histochemical reactions is to localize specific cell structures that express a given biochemical property.

BIostatISTICS, HISTOLOGY AND HISTOCHEMISTRY

Unit-I

BIostatISTICS

- 1.1 Data and its type: Discrete and continuous data, Frequency and non-frequency data,
- 1.2 Collection of data: Primary and secondary data
- 1.3 Presentation of data: pie chart, bar graphs, histograms
- 1.4 Measurement of central tendency: Mean, Median, Mode
- 1.5 Measurement of dispersion: Standard Deviation, Standard error, coefficient of variation

Unit-II

HISTOLOGY AND HISTOCHEMISTRY

- 1.1 Histology and Histochemistry - Introduction and definition
- 1.2 Fixative used in histology, Preparation and Application.
- 1.3 Tissue processing, Fixation, Dehydration, Hydration, Clearing, Embedding.
- 1.4 Staining Principle and Application : Eosin, Hematoxylin, PAS, Mercury bromophenol blue, Ninhydrin- Schiff, Sudan, Methyl green pyronin-Y, Feulgen reaction

Course Outcomes: It will enable the students to apply software in designing of experiment and its application in interpretation of research. It will enable the students to apply statistical concept in the field of public health.

The knowledge of histology will enable the students to differentiate between normal and pathological state of the tissues or organs. Medical conditions like cancer and detection of multi-organ diseases may be easily report with sound knowledge of/ histology.

The histochemistry will enable the students staining techniques for carbohydrates, proteins and lipid and other macromolecules as well. This will help them to locate the desired molecules in the cell.

MICROBIOLOGY, BIOTECHNOLOGY, BIOSTATISTICS, HISTOLOGY AND HISTOCHEMISTRY

Unit-I

MICROBIOLOGY

- 1.1 List at least two bacterial species to be Gram-positive and two species to be Gram-negative
- 1.2 Stain the milk/sputum bacterial smear prepared by using Gram staining technique
- 1.3 Recognize selected properties of bacterial colonies on agar plates.
- 1.4 List several common microbial infections of the skin
- 1.5 List several common microbial infections of the gastrointestinal tract.
- 1.6 List several common infections of the respiratory tract.
- 1.7 List common microbial pathogens and diseases of the genitourinary tract.

Unit-II

BIOTECHNOLOGY

- 1.1 Principle of PCR, types and applications
- 1.2 Principle of DNA, recombinant technology and applications

Unit-III

BIOSTATISTICS

- 1.1 Calculate the mean length (cm) of 10 *Tilapia* fishes
- 1.2 Calculate the median of number of dorsal fin rays of 10 (even) different fishes
- 1.3 Calculate the median of number of dorsal fin rays of 07 (odd) different fishes
- 1.4 Calculate the mode of dorsal fin rays in 20 *Rohu* fishes
- 1.5 Calculate the SD of length (cm) of 10 different fishes
- 1.6 Calculate the SE of length (cm) of 10 different fishes

Unit-IV

HISTOLOGY AND HISTOCHEMISTRY

- 1.1 Preparation of Alcoholic grade
- 1.2 Fixation of tissue: Dehydration and Hydration
- 1.3 Demonstration of Counter Staining: Eosin, Hematoxylin

Recommended books

Microbiology

1. Madigan, M.T., Martinko, J.M., Dunlap, P.V., Clark, D.P., 2011. Brock Biology of Microorganisms. 13th edition. Pearson Education Inc.
2. Stanier, R.Y., Ingraham, J.L., Wheelis, M.L., Painter, P.R., 1987. General Microbiology. Fifth edition, MacMillan.
3. Talaro, K.P., Chess, B., 2011. Foundation in Microbiology. 8th edition. McGraw-Hill
4. Agrios, G.N., 1988. Plant Pathology, Academic Press, London.
5. Lucas, John, A., 1998. Plant Pathology and Plant Pathogens, Wiley-Blackwell, CRC Press.
6. Tauro P., Kapoor K.K., Yadav K.S., 1996, An Introduction to Microbiology, Wiley Eastern Limited.
7. Frazier W.C. and Westroff D.C. (1988): Fourth Edition, Food Microbiology, McGraw Hill INC.
8. Madigan M.T., Martinko J.M., Dunlop P.V. and Clark D.P. (2011). Brock Biology of Microorganisms. 13th Edition, Pearson Education Inc.
9. Stanier R.Y., Ingraham J.L., Wheelis M.L. and Painter P.R. (1987). General Microbiology. Fifth Edition, MacMillan.
10. Willey J.M., Sherwood L. and Woolverton C.J. (2010). Prescott's Microbiology. 8th Edition, McGraw-Hill.
11. David M. Sylvia: Principles and Applications of Soil Microbiology, Prentice Hall, 1998.

Biotechnology

1. David M. Sylvia: Principles and Applications of Soil Microbiology, Prentice Hall, 1998.
2. T.A. Brown : Genomes, 2nd Editions, Oxford Willey-Liss, 2002.
3. Suzy Hill: Biotechnology, from sciences to applications
4. Dubey, R.C. (2006) A Text Book of Biotechnology. S. Chand & Co. Ltd. , New Delhi

Biostatistics

1. Fundamental of biostatistics: Khan and Khanam
2. W.W. (2012) Biostatistics: A Foundation for Analysis in Health Sciences (10th edition) John Wiley.
3. Milton, J.S. & Tsokos, J.O. (1992) Statistical Methods in the Biological and Health Sciences (2nd edition) McGraw Hill.
5. Zar, J.H. (2013) Biostatistical Analysis (5th edition) Pearson.

Histology and Histochemistry

1. Pearse A.G.E. - Histochemistry – Theoretical and Applied. vols I – III Churchill
2. Bancroft J.D. & Stevens A. – Theory and Practice of Histological techniques. 4th edn. Churchill Livingstone 1996
3. Barka T. & Anderson P.J. – Histochemistry, Theory Practice and Bibliography.
4. Harper and Raw 1965
5. Sharma A.K. & Sharma A. – Chromosome Techniques. Theory and Practice. 3rd edn. Butterworths 1980
6. Copenhaver W.M. – Bailey's Text Book of Histology. Willian and Wilkins/ Scientific Book Agency Indian edn. 1964
7. Carleton H.M. & Short R.M.D. – Schafer's Essential of Histology: Descriptive and Practical. 16th edn. Longmans Green 1954
8. Verma G.P. – Fundamentals of Histology. New Age 2001

Course Contents: The Head of the Department will divide all the students of the class into different groups and each group will be allotted topics from different field of Zoology Students will have to work under the supervision of a teacher of the department. Each and every student has to submit electronically typed hardbound dissertation along with the raw data on a week before the examination.

Steps of Evaluation:

1. Group discussion will be held within each group under the chairmanship of Head of the Department at the department level.
2. Presentation and the report submitted by the students will be evaluated by one external member and one internal member.

External Member will be appointed by the Head of Department from the following:

1. Permanent professors working in the postgraduate department of the university or other colleges.

OR

2. Retired Professor/Associate Professor/Assistant Professor of the university.

Evaluation Process:

Evaluation of the dissertation work may be as per the following guidelines:

- Dissertation -50 marks
- PPT Presentation -25 marks
- Viva- Voce -25 marks

Course objective: This paper incorporates some interesting topics which would help a person to understand trivial specimens of animal world with special attention on cytology and evolution of organisms.

Unit –I Animal kingdom

1. An overview of animal kingdom
2. General concept of non chordates up to phylum only
3. Classification of chordates up to class only
4. Prototheria & Metatheria

Unit –II Animal Cell

1. Cell (Prokaryotes & Eukaryotes)
2. Different organelles of Cell
3. Mitosis
4. Meiosis

Unit- III Evolution:

1. Origin of life
2. Theories of evolution (Lamarckism and Darwinism)
3. Homologus and analogous organs
4. Adaptive modification (Beaks & Claws of birds)
5. Convergent and Divergent evolution

Course outcome: This is an endeavor to give an initial input of Zoology to deal with some complex topics to be taught in higher semesters. Some interesting topics of cell biology and evolution will unfold many questions of animal world to the students within the discipline.

1. Study of Museum Specimens:

Sycon, Aurelia, Ascaris, Tapeworm, Prawn, Grasshopper, *Pila*, Octopus, Starfish, Balanoglossus, *Labeo*, *Scoliodon*, Frog, Lizzard, Cobra, Pigeon, Bat, Squirrel

2. Study of permanent Slides:

Mitosis, Meiosis, Amoeba, Paramecium, Hydra, Obelia, Leishmania

3. Evolution:

Homologus and Analogus organs, Adaptive modifications in beaks and claws of birds

Recommended Books:

Chordates

1. Kotpal, Agarwal & Khetrapal: Modern Textbook of Zoology: Invertebrate (Rastogi publication)
2. R. L.Kotpal: Invertebrate series: Protozoa to Minor phyla (Rastogi publication)
3. R.L.Kotpal: Modern textbook of Zoology: Vertebrate
4. Young, J.Z: Life of Vertebrates (Oxford University Press)
5. Hildebrand: Analysis of vertebrates Structure (Wiley)
6. E.L. Jordan & Dr. P.S. Verma: Chordate Zoology (S. Chand Publication)
7. Alexander, R.M.: The Chordates (Cambridge University Press)
8. Monaith, A. R.: The Chordates (Cambridge University Press)
9. Waterman, A. J.: Chordates: Structure and Function (Mac Millan Co.)
10. Nigam: Biology of Chordates (1997, S Chand)
11. Parker & Haswell: Text Book of Zoology, Vol. II (2005, Macmillan)

Cell biology and molecular biology

1. Karp, G. (2010). Cell and Molecular Biology: Concepts and Experiments. VI Edition. John Wiley and Sons. Inc.
2. De Robertis, E.D.P. and De Robertis, E.M.F. (2006). Cell and Molecular Biology. VIIIEdition. Lippincott Williams and Wilkins, Philadelphia.
3. Cooper, G.M. and Hausman, R.E. (2009). The Cell: A Molecular Approach. V Edition. ASM Press and Sunderland, Washington, D.C.; Sinauer Associates, MA.
4. Molecular Biology of the Gene - Watson, J.D et al (Benzamin / Commings)
5. Lehninger Principles of biochemistry: Cox & Nelson, MacMillan & Freeman, USA
6. Cell Biology- C.B. Pawar (Himalaya publishing house)

Evolution

1. Strik Berger, M.W. Evolution, Jones & Bartett. Publishers, Boston, London
2. An Introduction to Paleontology –A.P.Tyagi (S.Chand & Com.LTD)
3. Hall and Hallgrimsson: Strickberger's Evolution (2008, Jones and Bartlett)
4. Moody: Introduction to Evolution (1978, Kalyani).
5. Rastogi: Organic Evolution (2007, Kedarnath & Ramnath)
6. Evolution- Organic evolution Veer Bala Rastogi, MedTech Science Press
7. Futuyma: Evolutionary Biology (2005, Sinauer)

Practical Manual:

1. A Manual of Practical Zoology Invertebrates: S. Chand, Harnam Singh & Dr. P.S. Hemne
2. Practical Zoology Invertebrate and Vertebrate: S.S. Lal, Rastogi

Course Objectives: This paper has been prepared to include admixture of some important aspects zoology to provide the students maximum exposure of this subject within short span of time. The some topics of physiology, anatomy and animal behavior will infuse a comprehensive knowledge to increase interest in the course. Especially understanding of behavior is the highlight of this paper.

Unit –I Human Physiology

1. Digestion and absorption of dietary molecules
2. Respiration, Mechanism of Breathing
3. Circulation, Transport of O₂ & CO₂
4. Excretion, urine formation
5. Nervous system, Nerve impulse conduction (Myelinated and Non-Myelinated)

Unit –II Anatomy

1. Digestive System
2. Heart
3. Kidney
4. Structure of eye

Unit- III Animal Behavior

1. Innate and Learned Behavior
2. Social Behavior in Insects (Honey Bee and Termites)
3. Parental Care in Fish and Amphibia.

Course Outcomes: All the units of this paper will bring about a necessary boost and useful information about the life science. The physiology and anatomy are closely sections which give an opportunity to the students to understand the major functions of human body. The learning of animal behavior will be able to understand its scientific retrospect rather than social context that the people conjure up at any point of time.

1. Biochemical Test: Glucose, Urea, Protein (Urine Sample)
2. Study of Permanent Histological Slides: Kidney, Liver, Pancreas, Intestine.
3. Model- Eye, Heart, Kidney.
4. Study of museum specimen showing Parental care in Fish and Amphibia.

Recommended Books:

Human Physiology and Anatomy

1. Review of Medical physiology, Ganong (Lange)
2. Animal Physiology - Neilsen (Cambridge)
3. Animal Physiology - Eckert, R. (W. H. Freeman)
4. Review of Medical physiology, Ganong (Lange)
5. Reproductive Physiology - (Nalbandov, A.V.)
6. General & Comparative Physiology - Hoar (Prentics Hall)
7. Animal Physiology - Neilsen (Cambridge)
8. Comparative Animal Physiology - Progser (Satish Book Enterprise)
9. Textbook of Applied ANATOMY & PHYSIOLOGY- Surabhi Garg
10. A practical book of Human Anatomy and Physiology (D S Mandlik, S K Mandlik)

Animal Behaviour

1. An introduction to animal behavior: Aubrey Manning and Dawkins
2. Animal Behavior: Reena Mathur, (Rastogi Publication)
3. Animal behavior: VK Agarwal
4. Animal behavior: Natrajan and Arumugam
5. Alcock: Animal Behaviour: An evolutionary approach (9 ed. 2009, Sinauer)
6. Lorenz: The foundation of ethology (1981, Springer)
7. Manning & Dawkins: An Introduction to Animal Behaviour (5th ed. 1998, Cambridge).
8. Mcfarland: Animal Behaviour, Psychology, Ethology and Evolution (1985, Pitman).
9. Scott: Essential Animal Behaviour (2005, Blackwell)
10. Slater: An introduction to ethology (1985, Cambridge)

Course Objectives: This paper throws light on the topics of common interest as the questions about hormones and related disorders always perturb the people suffering from diabetes and thyroid problem. The developmental aspect dealing with gametogenesis and cleavage will create additional interest in the students. The economic zoology has been included to create skill development and self employment.

Unit –I Endocrinology

1. Histology of Endocrine glands and their Secretions and functions:
Pituitary, Thyroid, Pancreas (Islets of langerhans), Adrenal and Gonads

Unit –II Developmental Biology

1. Gametogenesis (oogenesis and spermatogenesis)
2. Types of eggs and Cleavage pattern
3. Development of three germ layers (general)

Unit- III Economic Zoology

1. Sericulture
2. Apiculture
3. Lac Culture
4. Pisciculture

Course Outcomes: The students will be equipped with the knowledge of some very common but important aspect of body physiology, embryology and economic zoology. This will enable the students to proceed further with diploma course, especially in the field of silk and honey production and to develop industry of considerable economy as being witnessed these days.

1. Demonstration of Endocrine Glands in Rat.
2. Study of Permanent Histological slide of Endocrine glands
3. W.M. of Chick embryo (24hrs, 48hrs, 72hrs, 96hrs.)
4. Life cycle of Silk worm, Lac worm, Honey bee
5. Economic importance:
 - Honey bee
 - Lac insect
 - Silk worm
 - Common carps

Recommended Books:

Endocrinology

1. Endocrinology - Hadley.
2. E.J.W. Barrington – General & Comparative Endocrinology, Oxford, Clarendon Press.
3. P.J. Bentley, Comparative Vertebrate Endocrinology, Cambridge University Press.
4. R.H. Williams - TextBook of Endocrinology. W.B. Saunders.
5. C.R. Martin - Endocrine Physiology, oxford.
6. A Gorbman et al. Comparative Endocrinology. John Willey & Sons

Development biology

1. Introduction to Embryology - Balinsky (CBS College publishers)
2. Developmental Biology - Subramaniyan, T (Narosa publishing House)
3. Developmental Biology - Gilbert
4. Chordate Embryology: Verma & Agarwal: S.Chand & Com. LTD
5. BIOS instant notes Developmental biology
6. Wolpert: Principles of Development (3rd ed. 2007, Oxford)

Economic Zoology

1. Shukla and Upadhyaya: Economic Zoology (Rastogi Publishers, 1999-2000)
2. Shrivastava: Test book of Applied Entomology, Vol. I &II (Kalyani Publishers, 1991)
3. Mani: Insects, NBT, India, 2006.
4. Jabde: Text Book of Applied Zoology: Vermiculture, Apiculture, Sericulture, Lac culture,
5. Agricultural Pests and their Control, 2005 Publisher Vedams eBooks (P) Ltd. New Delhi

Course Objectives: The lecture-discussion part is conceptualized with the aim that students are taught the basics of immunology so as to develop understanding of the subject such as “how does the immune system work”?

The concept of biomolecules is fundamental in understanding almost all sections of Zoology and especially physiology, cytology and molecular biology are based upon the concept of biochemistry.

The classical genetics incorporates chromosomal level inheritance with concept of genes and alleles. It explains how Mendel's particulate mechanism differed from blending theory of inheritance.

Unit –I Immunology

1. Introduction and scope of Immunology
2. Types of Lymphoid organs
3. Antigens and Antibodies interaction

Unit –II Biochemistry

1. Structure of DNA and RNA
2. Structure and function of the following :
 - (a) Carbohydrate
 - (b) Protein
 - (c) Lipid
3. Vitamin- classification, sources, function and deficiency
4. Enzyme- classification with example

Unit- III Genetics

1. Mendel's Law
2. Mutation
 - (a) Gene mutation
 - (b) Chromosomal aberration
3. Interaction of genes
 - (a) Interaction between allelic genes
 - (i) Complete dominance
 - (ii) Incomplete dominance
 - (iii) Co-dominance
 - (b) Interaction between non-allelic genes
 - (i) Dominant epistasis
 - (ii) Recessive epistasis
 - (iii) Duplicate genes

Course Outcomes: Upon completion of the course students have a sound understanding of the essential elements of the immune system, preparing them to engage further in this rapidly evolving field. It will encourage the students to explore designing new vaccines and other immunotherapeutic strategies.

The topics of biochemistry and tools & techniques will increase the horizon of knowledge in the students. Especially quantitative estimation of biochemical parameters as application the instruments will be quite handy in dealing with research assignment.

Classical genetics provide many insights into inherited traits and elucidate many inherited human disorders that are known to follow Mendel's law of inheritance and are useful to explain the appearance of diseases within families.

Zoology
Minor from Discipline

Practical
MN-04

SEM- VII

FM: 25, PM: 10

1. Preparation of blood film and identification of blood cells.
2. Biochemical test for
 - (a) Carbohydrate (Benedict)
 - (b) Protein (Biuret test)
 - (c) Lipid (Solubility)
3. Study of phenotypic trait of *Drosophila*
4. Model of DNA, RNA, Antigen, Antibodies.

Recommended Books:

Immunology

1. Kuby - Immunology W.H. Freeman, USA.
2. BIOS Instant Notes in Immunology
3. Kindit, T.J., Golds by R.A., Osborne, B.A. and Kuby, J (2006). Immunology, VI Edition. W.H. Freeman and Company.
4. David, M., Jonathan, B., David, R.B. and Ivan R. (2006). Immunology, VII Edition, Mosby, Elsevier Publication.
5. Abbas, K. Abul and Lechtman H. Andrew (2003) Cellular and Molecular Immunology. V edition. Saunders Publication

Biochemistry

1. Cox, M.M and Nelson, D.L. (2008). Lehninger Principles of Biochemistry, V Edition, W.H. Freeman and Co., New York.
2. Berg, J.M., Tymoczko, J.L. and Stryer, L. (2007). Biochemistry, VI Ed, W.H. Freeman and Co., New York.
3. Murray, R.K., Bender, D.A., Botham, K.M., Kennelly, P.J., Rodwell, V.W. and Well, P.A. (2009). Harper's Biochemistry (25th ed. 2000, Appleton & Lange)
4. Illustrated Biochemistry, XXVIII Edition, International Edition, The McGraw-Hill Companies Inc.
5. Hames, B.D. and Hooper, N.M. (2000). Instant Notes in Biochemistry, II Edition, BIOS Scientific Publishers Ltd., U.K.
6. Boyer: Concepts in Biochemistry (3rd ed. 2006, Brooks/Cole)

Genetics

1. Molecular biology of Gene: Watson et al., Pearson Publication, USA
2. Strickberger's Genetics, Prinitis Hall of India (PHI), Delhi
3. Principles of Genetics: Snustad & Simmons, John Wiley & Sons, USA
4. Genetics: PK Gupta, Rastogi Publication, New Delhi.
5. Genetics: Benjamin A pierce

COMPUTER APPLICATIONS AND RESEARCH METHODOLOGY

Course Objectives: The chief objectives of imparting education to the students are creation of job and to provide opportunity for earnings so as to lead a respectful and decent life. This course is meant to impart knowledge to students on the most important skill which is required in this era for any scientific worker. Particularly becoming conversant with computer handling can enhance the chances of self employment and also enable the students to make use of this indispensable acumen in other fields of Zoology like: computational biology, molecular biology, bioinformatics etc.

Since the eligible students will undertake Ph.D. course they must be equipped with the essentials of beginning research work. The contents of research methodology will certainly help them to develop research acumen for successfully completing their assigned synopsis.

Computer Science and Bioinformatics:

Unit-I:

- 1.1 Computer and its application in life sciences
- 1.2 Introduction of MS office – Word, Excel and Power point

Unit-II:

Digital library for life sciences (NCBI, Pub Med)

Unit-III

- 3.1 Bioinformatics and its application in life sciences
- 3.2 bioinformatics tools- BLAST, Clustal W, KEGG
- 3.3 Concepts of digital Library.

Research Methodology

Unit-I

- 1.1 Collection and grouping of data
- 1.2 Research design
- 1.3 Statistical analysis (t-test & ANOVA)

Unit-II

- 2.1 Identification and formation of research problem (Hypothesis)
- 2.2 Research design
- 2.3 Research proposal writing
- 2.4 Research article writing

Unit-III

- 3.1 Ideas of Scientific Journals (SJI, ISBBN, ISSN, Thompson, Reuter Index, Impact factor)
- 3.2 A brief idea of different Research Agencies for funding process

Course Outcomes:

This course is meant to impart knowledge to students on the most important skill which is required in this era for any scientific worker. The course is designed in such a way that the students get the confidence to use computer programs for the daily design of experiments, data collection, and analysis of results. The mandatory practical exercises in the available state of computer laboratory in the campus as ICT Lab. will benefit students to learn all that they require to use their computer for the study of science.

Since research methodology is based upon basic knowledge of computers, therefore these two will have cumulative effect on effective grasping and solution of research work undertaken

Hematology and applications of important instruments:

Course Objectives: To understand the patho-physiology of any disease requires a comprehensive understanding of in-depth knowledge in the field of hematology. The detail studies of blood composition and related diseases aim at providing very trivial information about the routine blood tests of the persons referred by the physician. Hence it will be quite handy in research and in joining pathological laboratories.

Hematology:

Unit-I

- 1.1 Scope, prospects and history of Hematology.
- 1.2 Blood Composition: Mammalian Blood corpuscular composition of Human beings
 - 1.2.1 Erythrocytes (RBC)
 - 1.2.2 Leucocytes (WBC)-types, structure and function and Blood platelets.

Unit-II

- 2.1 Haemopoiesis-process, stem cells, haemopoietic tissue
- 2.2 Structure and function of bone marrow: Leucocytosis and Leucopaenia

Unit-III

- 3.1 Composition of blood plasma in man
- 3.2 Anemia-diagnosis and therapy: Iron metabolism and iron deficiency anemia, megaloblastic anemia, pernicious anemia, aplastic anemia, sickle cell anemia, thalassemia, erythroblastosis fetalis

Unit-IV

- 4.1 Abnormal blood types: Polycythemia, Methamoglobinemia and its causes
- 4.2 Abnormal Haemoglobin, Spherocytosis, Megakaryopoiesis & Platelets disorder, Hemophilia A, and Hemophilia B
- 4.3 Blood transfusion and blood groups(ABO system and Bombay system)
- 4.4 Blood coagulation and Hemostasis, Blood coagulation inhibitors

Applications of important instruments:

Unit-I

- 1.1 Serum analysis: of
 - 1.1.1 Glucose
 - 1.1.2 Urea acid,
 - 1.1.3 Liver function test (SGPT & SGOT)

Unit-II

- 2.1 Auto-analyzer: Basic components and result finding process
- 2.2 ELISA technique: Theory, Kit study, ELISA plate reader
- 2.3 Spectroscopy: Principle and use.

Unit-III

- 3.1 Medical Imaging techniques using photographs and reports:
 - 3.1.1 X-Ray
 - 3.1.2 PET

Course Outcomes: Hematology dealing with the blood composition incorporates the most fundamental biochemical tests of several diseases and diagnosis of the same by examining the referred parameters associated with them. Such students can contribute as specialized teaching faculty after acquiring the postgraduate degree with this special paper. Such students may also undertake MLT course with the acquired knowledge in this field. The episodes of pandemic have further underscored the significance of hematological tests.

1. Use of MS Office
2. PPT preparation and presentation
3. Applications Digital library- NCBI, Pub Med
4. Applications of BLAST, Clustal W, KEGG in research
5. Calculations of data using t-test & ANOVA
6. Preparation of synopsis- one example to explain
7. List preparation of scientific journals in life sciences
8. Preparation of blood film
9. RBC & WBC Count
10. Hb% estimation
11. Quantitative estimation of Protein by colorimeter
12. Preparation of Phosphate buffer
13. Sample preparation for Scanning Electron Microscope

Recommended Books:

Tools and techniques

1. Roymahoney: Laboratory Techniques in Zoology (1966, Butterworths).
2. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,)
3. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill- Livingstones)
4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill)
5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)

Biostatistics

1. Fundamental of biostatistics: Khan and Khanam
2. W.W. (2012) Biostatistics: A Foundation for Analysis in Health Sciences (10th edition) John Wiley.
3. Milton, J.S. & Tsokos, J.O. (1992) Statistical Methods in the Biological and Health Sciences (2nd edition) McGraw Hill.
5. Zar, J.H. (2013) Biostatistical Analysis (5th edition) Pearson.

Research Methodology

1. Research Methodology – Text and Cases with SPSS Applications, by Dr S.L. Gupta and Hitesh Gupta, International Book House Pvt Ltd
2. Methodology of Research in Sciences, by O.R. Krishnaswami, Himalaya Publishing House
3. Research Methodology, Methods and Techniques by C.R Kothari
4. Research Methodology by Dr Vijay Upagude and Dr Arvind Shende
5. Gupta, Santhosh Research Methodology and Statistical Techniques, Deep and Deep.
6. Dawson, C. (2002). Practical research methods. UBS Publishers, New Delhi.
7. Stapleton, P., Yondeowei, A., Mukanyange, J., Houten, H. (1995). Scientific writing for agricultural research scientists – a training reference manual. West Africa Rice Development Association, Hong Kong.
8. Ruzin, S.E. (1999). Plant micro technique and microscopy. Oxford University Press, New York, U.S.A

Hematology

1. Hematology: basic principles and practice, 7th ed. – by Ronald Hoffman, Edward J. Benz
 2. Harrison's Hematology and oncology, 3rd ed, by Dan L. Longo (McGraw-Hill)
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SEMESTER-VIII

GUIDELINE AND SYLLABUS FOR 4-YEAR UG DEGREE (HONOURS WITH RESEARCH)

Eligibility for Students:

Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year. They should do a research project or dissertation under the guidance of a faculty member of the University/College. The research project/dissertation will be in the major discipline. The students, who secure 160 credits, including 12 credits from a research project/dissertation, are awarded UG Degree (Honours with Research).

Infrastructure Requirement for offering this Course:

The Departments offering a 4-year UG Degree (Honours with Research) must have the required infrastructure such as the laboratory, equipment's, glass wares, chemicals, class work materials, library, access to journals and computer lab facilities to carry out the research work, and at least two permanent faculty members who are recognized as Ph.D. supervisors. The Departments already recognized for conducting the Ph.D. programme may conduct a 4- year UG Degree (Honours with Research) without obtaining any approval from the affiliating University.

Semester	Course Category	Code	Papers	Credits
Eighth Semester	Research courses	RC-01	Research Methodology	4
	Research Project	RC-02	Research Project or Dissertation	8
			Total Credits	12

RC-ZOO-01 (RESEARCH METHODOLOGY)

- *Total Credits: -04*
- **Total teaching Hours: 60**
- *Total Marks: 100*
- **Semester Internal Examination: 25 Marks**
- *End Semester University Examination: 75 Marks*

Objective:

• The course aims to inculcate research aptitude among the learners and equip them with knowledge and skills required to successfully undertake various steps in the research process.

Course Outcomes: After completion of the course, learners will be able to:

- Analyze research concepts, its types and steps in the research process.
- Formulate research problems and assimilate various types of research design.
- Create scales, sampling techniques and design data collection tools.
- Recognize various techniques of data analysis and interpretation.
- Prepare a complete research report in appropriate format.

Unit-1: Introduction of Research

Research: Meaning, Definition, Nature, Scope, Significance, Steps Types and Methods of Research.
Characteristics of Goods Research

Unit-II: Research Process

Defining Research Problem, Title Formulation; Setting of Hypothesis, Research Design - Exploratory, Descriptive and Experimental Research Designs.

Unit-III: Tools and techniques

PCR, Electrophoresis, Spectrophotometer, Colorimeter, RIA, ELISA.

Unit-IV: Data

Data, types of data, Methods of Collection of Primary and Secondary Data. Process of Questionnaire Design; Processing of Data-Editing, Coding, Classification and Tabulation.

Unit-V: Sample and Sampling

Meaning of Sample, Sampling Design, Criteria of Selecting a Sampling Procedure, Characteristics of Good Sample Design.

Unit-VI: Analysis of Data

Chi-square test, t-test, ANOVA, Correlation, Regression

Unit-VII: Report Writing

Meaning and Types of Report, Steps of Report Writing, and Characteristics of a good report, Reference and bibliography.

Recommended Books:

Tools and techniques

6. Roymahoney: Laboratory Techniques in Zoology (1966, Butterworths).
7. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,)
8. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones)
9. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill)
10. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)

Biostatistics

6. Fundamental of biostatistics: Khan and Khanam
7. W.W. (2012) Biostatistics: A Foundation for Analysis in Health Sciences (10th edition) John Wiley.
8. Milton, J.S. & Tsokos, J.O. (1992) Statistical Methods in the Biological and Health Sciences (2nd edition) McGraw Hill.
10. Zar, J.H. (2013) Biostatistical Analysis (5th edition) Pearson.

Research Methodology

9. Research Methodology – Text and Cases with SPSS Applications, by Dr S.L. Gupta and Hitesh Gupta, International Book House Pvt Ltd
10. Methodology of Research in Sciences, by O.R. Krishnaswami, Himalaya Publishing House
11. Research Methodology, Methods and Techniques by C.R Kothari
12. Research Methodology by Dr Vijay Upagude and Dr Arvind Shende
13. Gupta, Santhosh Research Methodology and Statistical Techniques, Deep and Deep.
14. Dawson, C. (2002). Practical research methods. UBS Publishers, New Delhi.
15. Stapleton, P., Yondeowei, A., Mukanyange, J., Houten, H. (1995). Scientific writing for agricultural research scientists – a training reference manual. West Africa Rice Development Association, Hong Kong.
16. Ruzin, S.E. (1999). Plant micro technique and microscopy. Oxford University Press, New York, U.S.A

RC-ZOO-02 (RESEARCH PROJECT OR DISSERTATION)

- **Total Credits: -08**
- **Total Marks: 200**

Objectives: The objectives of the course is to facilitate students to carry out extensive research and develop as self-guided learning and analytical skills through problem and gap identification, development of research methodology, interpretation of findings and presentation of results.

Course Outcomes: After completion of the course, the learners will be able to:

- Gain in-depth knowledge in the major field of study.
- Design and justify research methodology.
- Utilize appropriate research methodology for research.
- Analyze the data and draws conclusions accordingly.

Course Contents:

Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year.

The HOD of the department has to allot supervisor to the students from among the permanent faculty members who have Ph.D. degree, after that the students have to select a research problem with the help of the supervisor and they have to submit a summary or research proposal to the department. Thereafter, the HOD of the department will organize a meeting of the Departmental Research Committee and after the presentation of the student the committee will approve or reject his/her synopsis/research proposal. Students will start their research work after getting approval from the department research committee.

At the end of the semester the student has to submit the project thesis to the department and after that the department will send that thesis to the external member, after receiving the report from the external member the department will conduct an open viva voce examination for the students in the presence of the external member as well as the internal member.

The Department Research Committee will be constituted from the following persons:

1. HOD of the Department–Chairmen
2. Dean, Faculty of Science- External member
3. HOD, University Department- External member
4. At list two permanent faculty member of the department (Nominated by HOD)- Members

External members can be any of the following:

1. Permanent professors working in the postgraduate department of the university or other colleges who have the qualification to become PhD supervisors.

OR

2. Retired Professor/Associate Professor/Assistant Professor of the university who has been supervise PhD scholar.

OR

3. Professor/Associate Professor/Assistant Professor of the outside university who has been supervise PhD scholar.

Note- Minimum three external examiner lists will be sent by HOD through concern college principal to controller of examination, SKMU, Dumka for final approval. In that list priority will be given as per order mentioned above.

The project thesis will evaluate under the following heads:

- Motivation for the choice of topic
- Project dissertation design
- Methodology and Content depth
- Results and Discussion
- Future Scope & References
- Participation in Internship programme with reputed organization
- Application of Research technique in Data collection
- Report Presentation
- Presentation style Broad

Guidelines for distribution of marks may be as follows or as appropriate:

- Assessment of project synopsis: 75 marks
- Assessment of project thesis: 100 marks
- Viva-voce: 25 marks